

Curvature Polarization Transport Gravity:

A Unified Geometric Framework for Cosmic Structure and Expansion

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Abstract

Curvature Polarization Transport Gravity (CPTG) is presented as a unified geometric framework in which ordinary baryonic matter sources nonlinear curvature response rather than requiring a separate dark halo component or a collection of unrelated cosmological fit parameters. The theory is organized around curvature polarization, which controls the local strength and structure of the gravitational response, and curvature transport, which carries organized curvature across dynamical and cosmological settings.

Within this framework, galaxy rotation support, outer-slope convergence, the Curvature-Weighted Structural Mode Index, dissociative cluster-merger lensing offsets, cosmic expansion, clock relaxation, growth response, weak-lensing response, baryon conversion, acoustic transport, and CMB-related comparison quantities are treated as reduced channels of the same geometric theory. The goal is not to redefine CPTG in the language of observational pipelines, but to show that a single native curvature branch can scale from bound structure to expansion while still producing the quantities needed for controlled comparison with galaxy, cluster, and cosmology-facing observations.

This paper introduces the core theory: the native geometric branch, the nonlinear field structure, the reduced galaxy and cluster limits, the structural diagnostics, the expansion and clock branch, the perturbation and lensing channels, and the amplitude and visibility gates used by the CMB comparison layer. CPTG is therefore developed here as a compact, scalable geometric theory of cosmic structure and expansion.

1 Framework Principle

This paper states the core CPTG theory: the native geometric branch, the nonlinear curvature-polarization field structure, the reduced galaxy and cluster limits, the structural diagnostics, the expansion and clock branch, the growth and lensing response, and the CMB-facing acoustic and visibility gates. The broader framework and supporting reproducibility resources are documented in Refs. [8, 12].

CPTG treats curvature response as a transported and polarized geometric field, not as a set of independently fitted sectors. The theory first defines native geometric quantities and only afterward converts them into the variables used by

observational likelihoods or published comparison summaries. The comparison hierarchy is

$$\begin{aligned} \text{CPTG-native geometric quantity} &\longrightarrow \text{comparison-layer observable} \\ &\longrightarrow \text{reported observational quantity.} \end{aligned} \quad (1)$$

The first layer is fixed by the geometric branch. The second layer expresses the native result in the variables used by a particular observational pipeline. The third layer is the reported comparison value, which may depend on a fiducial cosmology, likelihood convention, calibration choice, nuisance-parameter treatment, or compression scheme. This separation is essential: CPTG-native quantities are not fitted to the reported values, but converted into the appropriate comparison coordinates.

In this paper, cosmic structure means the hierarchy of reduced curvature-response channels that connect local and cosmological systems: galaxy-scale polarization in weak fields, cluster-merger transport in dynamical curvature fields, growth response in the large-scale matter distribution, and lensing response in projected curvature observables. These are not separate phenomenological sectors. They are scale reductions of the same transported and polarized curvature structure.

The geometric acceleration anchor is

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2}. \quad (2)$$

Numerically,

$$a_{\pi,0} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}. \quad (3)$$

The corresponding asymptotic Hubble scale is

$$H_{\infty} = \frac{a_{\pi,0}}{c}. \quad (4)$$

In conventional cosmological units this gives

$$H_{\infty} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (5)$$

The raw pi-branch present-epoch normalization follows from the late-branch weight

$$\mathcal{B}_{\pi} = 1 - \frac{1}{\pi}, \quad (6)$$

so that

$$H_0^{(\pi)} = \frac{H_{\infty}}{\sqrt{\mathcal{B}_{\pi}}}. \quad (7)$$

This gives

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (8)$$

The branch curvature scale is

$$R_C = 48\pi \frac{c^2}{a_{\pi,0}}. \quad (9)$$

Therefore,

$$R_C = 788.7823091477 \text{ Gpc}. \quad (10)$$

The quantities

$$a_{\pi,0}, \quad H_\infty, \quad H_0^{(\pi)}, \quad R_C \quad (11)$$

define the native geometric normalization of the branch. They are the starting point for comparison-layer conversions, not fit outputs from observational summaries.

2 Nonlinear Field Structure and Scalar Reductions

CPTG is a nonlinear geometric field theory. Its scalar equations are not independent postulates. They are closure laws obtained after reducing the full curvature-polarization and curvature-transport structure under the symmetries appropriate to a given comparison problem.

A schematic CPTG action may be written as

$$S_{\text{CPTG}} = \int d^4x \sqrt{-g} \left[\frac{c^4}{16\pi G} R + \mathcal{L}_{\text{pol}} + \mathcal{L}_{\text{trans}} + \mathcal{L}_{\text{m}} \right]. \quad (12)$$

Here,

$$\mathcal{L}_{\text{pol}} \quad (13)$$

represents the curvature-polarization sector,

$$\mathcal{L}_{\text{trans}} \quad (14)$$

represents the curvature-transport sector, and

$$\mathcal{L}_{\text{m}} \quad (15)$$

is the matter-source Lagrangian.

Variation with respect to the metric gives the schematic nonlinear field equation

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (16)$$

The tensors

$$\Pi_{\mu\nu}^{\text{pol}} \quad \text{and} \quad \Theta_{\mu\nu}^{\text{trans}} \quad (17)$$

are nonlinear geometric-sector contributions. They are not external matter components, dark-sector fluids, or adjustable correction terms. They encode how curvature polarizes and transports across scale.

The operational hierarchy is

$$\begin{aligned} \text{nonlinear CPTG field equation} &\longrightarrow \text{symmetry and scale reduction} \\ &\longrightarrow \text{scalar closure law} \\ &\longrightarrow \text{comparison-layer observable.} \end{aligned} \quad (18)$$

The scalar closure laws used in cosmology and structure calculations are therefore reduced forms of the same nonlinear field structure. They include background expansion, acoustic transport, clock relaxation, growth response, lensing response, primordial-abundance conversion, CMB comparison variables, and compressed large-scale-structure observables.

Representative reduced scalar quantities are

$$E(a)^2, \quad \mu(a), \quad \Sigma(a), \quad A_{\text{bandpower}}, \quad S_{8,\text{CMBL}}. \quad (19)$$

These quantities are comparison-layer reductions. They are useful because they allow the nonlinear CPTG field structure to be tested against observational summaries and likelihood coordinates, but they should not be identified with the full theory itself.

The distinction is essential. The nonlinear field equation defines the theory. The scalar laws provide the reduced closure needed to compare that theory with specific observables.

3 Galaxy-Scale Curvature Polarization Reduction

The galaxy sector is the bound, quasi-static reduction of the same nonlinear field structure introduced above. Disk-galaxy rotation curves provide the cleanest public test of this limit because the observed circular velocity can be compared directly with the acceleration field solved from the baryonic source [1, 2]. In this regime explicit time dependence is neglected, approximate rotational symmetry is used, and the field is represented by a scalar potential

$$\mathbf{g} = -\nabla\Phi, \quad g = |\mathbf{g}|. \quad (20)$$

The local galaxy reduction is controlled by a curvature-structure invariant of the form

$$C_g(r) = g(r)^2 + \kappa_g^2 \left(\frac{dg}{dr} \right)^2. \quad (21)$$

The first term measures field strength. The second term measures radial organization of the solved field. The invariant is therefore not a dark matter density profile. It is a diagnostic of how baryonic geometry sources a nonlinear curvature response.

In the galaxy limit, curvature polarization and the radial remnant of curvature transport combine into an implicit nonlinear acceleration equation. The reduced field equation is

$$g(r) = g_{\text{bar,CPTG}}(r) + \frac{\sqrt{a_\star g_{\text{bar,CPTG}}(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_\star)^{123/50}\right]^{11/50}}. \quad (22)$$

This equation is the galaxy-level engine of the theory. The total acceleration appears on both sides, so the field is solved rather than assigned. The first term is the baryonic source. The square-root term is the leading curvature-polarization response. The geometric term is the surviving radial transport remnant. The denominator suppresses the added curvature response in the high-acceleration regime, allowing compact inner systems to return toward the baryonic Newtonian limit.

For clarity, Eq. (22) can be read as

$$g_{\text{CPTG}}(r) = g_{\text{bar,CPTG}}(r) + \Delta g(r), \quad (23)$$

with

$$\Delta g(r) = [g_{\text{pol}}(r) + g_{\text{geom}}(r)] \Sigma(r), \quad (24)$$

where

$$g_{\text{pol}}(r) = \sqrt{a_\star g_{\text{bar,CPTG}}(r)}, \quad \Sigma(r) = \left[1 + (g(r)/a_\star)^{123/50}\right]^{-11/50}. \quad (25)$$

This decomposition separates the physics without changing the equation: baryonic sourcing, curvature polarization, geometric transport, and nonlinear suppression are all present in the same solved field.

Here $a_\star$ is a local structural acceleration scale, not the cosmological anchor $a_{\pi,0}$. CPTG links acceleration to structure, so different regimes can carry different acceleration scales while still belonging to the same curvature-response family. In the galaxy equations, $a_\star$ organizes the local polarization response; in the cosmic branch, $a_{\pi,0}$ sets the native expansion scale. The galaxy-limit transport remnant is

$$g_{\text{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad \eta = \frac{1}{2}. \quad (26)$$

The structural transport radius is derived from the same acceleration scale:

$$R_c = \frac{48\pi c^2}{a_\star}. \quad (27)$$

Equivalently, one may introduce the structural Hubble-equivalent scale

$$H_{\star} = \frac{16\pi}{3} \frac{a_{\star}}{c}, \quad (28)$$

so that

$$R_c = \frac{c}{H_{\star}} (16\pi)^2. \quad (29)$$

This is a galaxy-structure comparison scale. The native cosmological branch is still defined independently by $a_{\pi,0}$, H_{∞} , and $H_0^{(\pi)}$.

The baryonic input field is built once from the observed gas, disk, and bulge source components before the nonlinear equation is solved. When a tabulated component carries a sign, the signed-square convention preserves the source orientation:

$$Q(V) = V|V|. \quad (30)$$

For positive component velocities this reduces to the ordinary square. The CPTG source velocity is then

$$V_{\text{bar,CPTG}}^2(r) = \frac{27}{50} Q(V_{\text{gas}}(r)) + \frac{23}{50} Q(V_{\text{disk}}(r)) + \frac{16}{25} Q(V_{\text{bulge}}(r)), \quad (31)$$

and

$$g_{\text{bar,CPTG}}(r) = \frac{V_{\text{bar,CPTG}}^2(r)}{r}. \quad (32)$$

The gas–disk pair closes as

$$\frac{27}{50} + \frac{23}{50} = 1, \quad (33)$$

while the compact bulge source carries the geometric weight

$$\frac{16}{25} = \left(\frac{4}{5}\right)^2. \quad (34)$$

For compact notation below, g_{bar} denotes this CPTG baryonic source field unless a comparison source is explicitly named.

The circular velocity is recovered only after solving Eq. (22):

$$V(r) = \sqrt{g(r)r}. \quad (35)$$

Thus the galaxy rotation curve is a solved curvature-polarization response to the observed baryonic source, not a halo profile assigned to the galaxy and not a universal interpolation curve laid over the source.

This reduction gives the galaxy sector a definite logical order. First, the nonlinear acceleration equation defines the response to the baryonic source. Second, the

intermediate weak-field limit gives the baryonic Tully–Fisher scaling. Third, the solved field contains a curvature-organization observable, the Structural Mode N . Fourth, the far-outer continuation of the same solution defines the slope of convergence. These are not separate assumptions; they are successive readings of the same reduced galaxy equation.

The limiting behavior is immediate. For high acceleration,

$$g \gg a_\star \quad \Rightarrow \quad g \approx g_{\text{bar,CPTG}}, \quad (36)$$

so the Newtonian baryonic limit is recovered. For low acceleration,

$$g \ll a_\star \quad \Rightarrow \quad g \sim \sqrt{a_\star g_{\text{bar,CPTG}}} + g_{\text{geom}}(r), \quad (37)$$

so the leading enhancement is polarization dominated while the transport remnant remains structure dependent.

4 Galaxy Structure and the Baryonic Tully–Fisher Scaling

The first major galaxy-level consequence of the reduced equation is the baryonic Tully–Fisher scaling. This scaling is not imposed as a phenomenological rule. It appears when Eq. (22) is read in the intermediate low-acceleration interval where observed disk rotation curves are approximately flat.

In the intermediate low-acceleration regime relevant to observed flat rotation curves, suppose

$$g_{\text{bar}}(r) \ll a_\star, \quad g(r) \ll a_\star, \quad (38)$$

and suppose the geometric remnant is subdominant over the flat-curve interval:

$$|g_{\text{geom}}(r)| \ll \sqrt{a_\star g_{\text{bar}}(r)}. \quad (39)$$

Then Eq. (22) reduces to

$$g(r) \simeq \sqrt{a_\star g_{\text{bar}}(r)}. \quad (40)$$

For an outer baryonic source approximated by an effective enclosed mass,

$$g_{\text{bar}}(r) \simeq \frac{GM_{\text{eff}}}{r^2}, \quad g(r) = \frac{V^2(r)}{r}, \quad (41)$$

Eq. (40) gives

$$V_f^4 \simeq Ga_\star M_{\text{eff}}. \quad (42)$$

If the effective source mass is written as

$$M_{\text{eff}} = \Xi M_b, \quad (43)$$

then

$$V_f^4 \simeq G a_\star \Xi M_b. \quad (44)$$

The baryonic Tully–Fisher slope therefore appears as the leading curvature-polarization scaling of the galaxy reduction, not as an imposed interpolation law [3].

This result is a scaling law, not the full galaxy theory. It appears when the solved curvature-polarization response is viewed in the intermediate low-acceleration interval where observed disk rotation curves are approximately flat. CPTG allows residual structure around Eq. (44) because the field is solved from the detailed baryonic source rather than collapsed into a single universal acceleration prescription. The Tully–Fisher relation captures the leading amplitude scaling of the galaxy response. It does not exhaust the solved field. The remaining structure of the solution is what motivates the Structural Mode N .

5 Structural Mode N as a Curvature-Organization Observable

The reduced galaxy equation does more than reproduce a mean rotation-curve scaling. Once the baryonic source is supplied and the CPTG field is solved, the solution contains an internal organization scale. CPTG identifies this scale through the Curvature-Weighted Structural Mode Index, denoted by N . The purpose of N is to measure how curvature support is organized inside the solved galaxy field, not to relabel the galaxy by morphology or by a fitted halo parameter.

The dimensionless mode index is defined by

$$N = \frac{R}{\lambda}, \quad (45)$$

where R is the outer resolved size of the system and λ is the curvature-organization scale extracted from the solved acceleration field. In the reduced galaxy implementation one first constructs a curvature-weighted structural density,

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad (46)$$

and the corresponding organization weight,

$$w(r) = \left| \frac{dC_g}{dr} \right|. \quad (47)$$

The outer structural scale is then read from the curvature-weighted radial centroid,

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (48)$$

The lower limit $R/5$ isolates the resolved outer structure where curvature support is most diagnostic and avoids allowing the bright inner disk or bulge to dominate the organization scale.

This makes N a theory-side observable. It is not assigned from catalog morphology, Hubble type, color, stellar mass, surface brightness class, or a halo concentration. It is also not a free fitting label. If the baryonic source, distance, mass-to-light scaling, or structural branch is changed, the field must be solved again before N can be recomputed. The value of N therefore changes only through the curvature response of the solved system.

The physical content of N is the number of outer curvature-organization lengths that fit across the resolved galaxy scale. A small change in the detailed baryonic distribution can alter the curvature-weighted centroid, while a coherent family of galaxies can cluster around similar mode values even when their visual morphologies differ. This is why N is not merely another residual statistic. It converts the shape of the solved gravitational field into a compact geometric descriptor.

This is a central novelty of CPTG. Standard halo modeling can assign a concentration, scale radius, or profile family after fitting a dark component, and MOND-like descriptions can organize galaxies by acceleration scale. CPTG instead predicts that the baryon-sourced curvature field itself carries a measurable structural mode. The claim is not that galaxy classification is new; the claim is that a gravity theory can compute a structural organization index directly from the solved curvature response.

The index is falsifiable. If N is meaningful, mode-filtered galaxy subsets should exhibit coherent behavior in rotation-curve residuals, outer-slope continuation, distance-refinement tests, and structural acceleration scales. If the values scatter randomly under improved data, or if mode-selected subsets do not improve structural coherence relative to arbitrary grouping, the proposed curvature-organization interpretation would fail. Thus N gives CPTG a new kind of test: not only whether a curve can be matched, but whether the theory discovers hidden structural order in the galaxy population. Where N measures curvature organization across the resolved galaxy domain, the slope of convergence describes how the same solved field continues beyond that domain.

6 Slope of Convergence in the Far Outer Galaxy Limit

The CPTG slope of convergence is the far-outer counterpart to the finite-domain Structural Mode N . It is a separate theoretical consequence of the reduced galaxy equation and should not be merged with the baryonic Tully–Fisher scaling or with the measured flat portion of a galaxy rotation curve. The Tully–Fisher relation is an intermediate low-acceleration scaling. The slope of convergence is a far-outer continuation property of the solved CPTG field.

Define the local logarithmic velocity slope

$$s(r) = \frac{d \ln V}{d \ln r}. \quad (49)$$

When the solved CPTG profile is extended beyond the outermost measured radius under a fixed-outer-source continuation rule, the theory predicts approach toward the CPTG outer attractor

$$s_{\infty} = \frac{1}{4}. \quad (50)$$

Equivalently,

$$V(r) \propto r^{1/4}, \quad g(r) \propto r^{-1/2}. \quad (51)$$

This result is new theory content because it is not inserted as a halo slope, a fitted extrapolation law, or a MOND interpolation choice. It is the asymptotic behavior of the curvature-polarization solution after the observed disk is continued into the far outer weak-field regime. The physical claim is that galaxy outskirts should not drift arbitrarily once the baryonic source is fixed. They should converge toward a stable curvature-polarization pattern.

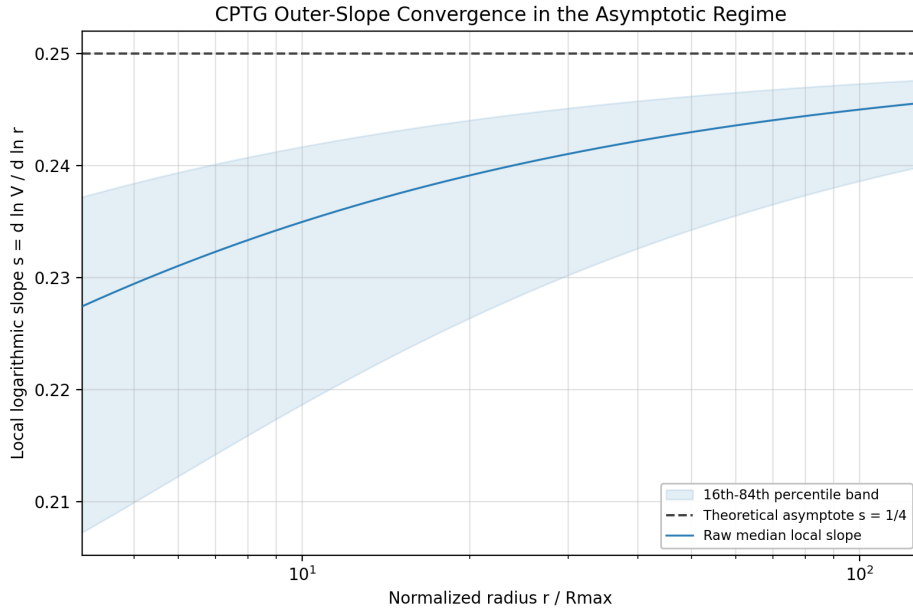


Figure 1: CPTG outer-slope convergence in the asymptotic extension regime. The solid curve shows the raw median local logarithmic slope $s = d \ln V / d \ln r$ as the solved CPTG rotation-curve behavior is extended beyond the observed SPARC rotation-curve domain under a fixed-outer-source continuation. The shaded band marks the central galaxy-to-galaxy spread, and the dashed line marks the theoretical outer-regime attractor $s = 1/4$. The visualization is not an additional fit; it displays the convergence behavior implied by the reduced galaxy equation.

The slope of convergence therefore provides a falsifiable outer-regime target. Extended HI maps, satellite tracers, weak-lensing galaxy outskirts, and future ultra-low-surface-brightness measurements can test whether solved CPTG profiles approach this predicted continuation behavior. Agreement would support the idea

that the galaxy outskirts are organized by curvature polarization; persistent, well-measured departures from the predicted convergence slope would stress or falsify this reduced galaxy branch.

7 Dissociative Cluster-Merger Transport Reduction

The same curvature-polarization and curvature-transport structure has a different reduction in dissociative cluster mergers. A cluster merger is dynamically anisotropic: collisional gas, galaxies, and lensing structure do not remain spatially coincident. The Bullet Cluster is the standard benchmark example of such gas–lensing separation [4].

In a reduced merger-plane description, the baryonic field is separated into gas and galaxy sectors,

$$\phi_{\text{bary}} = \phi_{\text{gas}} + \phi_{\text{gal}}. \quad (52)$$

The gas traces the collisional plasma, while the galaxy field traces the less-collisional baryonic concentrations. The galaxy-dominated geometry defines a curvature-organization background, and the full baryonic field defines a broad polarization-supported lensing background.

The effective lensing curvature in the transport-retaining reduction is

$$\mathcal{K}_{\text{lens}} = \nabla^2 \Phi - s \nabla \cdot [\Sigma^2 \mathbf{S}(\mathbf{S} \cdot \nabla) \Phi]. \quad (53)$$

The first term is the ordinary baryonic curvature source. The second term is the leading projected transport contribution along the reduced spatial transport direction $\mathbf{S}$. This term allows curvature support to be directionally redistributed in a merger environment.

The active transported component may be represented by a positive transported-curvature mode $T_+(x, y, t)$, with invariant

$$C_T = T_+^2 + \kappa_T^2 |\nabla T_+|^2. \quad (54)$$

A representative reduced transport equation is

$$\partial_t T_+ = D_{\text{trans}} \nabla^2 T_+ - \mathbf{u}_{\text{eff}} \cdot \nabla T_+ - \Gamma_{\text{trans}} T_+ + S_+[T_+] + \mathcal{S}_{\text{CPTG}}[T_+; \mathbf{u}_{\text{eff}}]. \quad (55)$$

Here D_{trans} describes spreading, $\mathbf{u}_{\text{eff}}$ is the merger-plane transport field, Γ_{trans} suppresses unsupported structure, S_+ is the positive reinforcement source, and $\mathcal{S}_{\text{CPTG}}$ is the curvature-support term.

A representative coherence-weighted support factor has the reduced form

$$\mathcal{S}_{\text{CPTG}} \propto \widehat{C}_T^{1/3} (1 + \mathcal{C}_{\text{coh}})^{1/3} \widehat{G}_T^{7/20} T_+, \quad (56)$$

where

$$\mathcal{C}_{\text{coh}} = \max \left(0, \widehat{\nabla C_T} \cdot \hat{\mathbf{u}}_{\text{eff}} \right)^2. \quad (57)$$

The observable reduced lensing field is then a polarization-supported background plus a transported-curvature contribution:

$$\Sigma_{\text{final}} = B_{\text{pol}} \Sigma_{\text{pol}} + \alpha_{T,\text{eff}} \Sigma_T. \quad (58)$$

This is not a hidden collisionless matter component. It is the projected curvature response generated by baryonic geometry, broad curvature polarization, and coherent directional transport.

The merger limit therefore complements the galaxy limit. In galaxies, curvature transport is coarse-grained into a radial remnant. In cluster mergers, transport remains explicit and can displace the lensing response relative to the gas. The same theory contains both behaviors because the role of transport depends on the dynamical symmetry of the system.

8 Theory Position Relative to MOND and Λ CDM

CPTG can reproduce MOND-like low-acceleration scaling in galaxies, but it is not a MOND interpolation-function model [5]. MOND introduces a characteristic acceleration scale and an interpolation law. CPTG instead obtains the square-root response as a reduced consequence of nonlinear curvature polarization:

$$g \sim \sqrt{a_{\star} g_{\text{bar}}} + g_{\text{geom}}(r). \quad (59)$$

The similarity is therefore phenomenological. The physical interpretation is different: MOND modifies the acceleration law, while CPTG computes a curvature response sourced by baryonic structure.

CPTG also differs from Λ CDM. In Λ CDM, galaxy and cluster anomalies are usually attributed to non-baryonic dark matter, often organized through halo profiles such as Navarro–Frenk–White forms [6, 7]. CPTG instead attributes the excess gravitational phenomenology to nonlinear geometric response. Galaxies probe the polarization-dominated limit; dissociative cluster mergers probe the polarization-plus-transport limit; cosmology-facing tests probe scalar reductions of the same branch.

The point of the comparison is not to deny the observations motivating MOND or Λ CDM. The point is to identify a different source for the same phenomena: baryonic geometry polarizes and transports curvature, rather than requiring either a universal interpolation rule or an independently assigned dark halo.

9 Immediate Structure-Sector Consequences

The galaxy and cluster reductions imply a set of direct structure-sector predictions.

First, compact high-acceleration systems should recover the Newtonian baryonic limit:

$$g \gg a_\star \quad \Rightarrow \quad g \approx g_{\text{bar}}. \quad (60)$$

Second, weak-field galaxies should show an enhanced curvature-polarization response with leading behavior

$$g \sim \sqrt{a_\star g_{\text{bar}}}. \quad (61)$$

Third, BTFR residuals should not be purely random. They should carry information about $a_\star$, the source-structure factor Ξ , the geometric transport remnant g_{geom} , and the structural mode index N . Fourth, deep outer rotation-curve measurements should begin to distinguish the observed intermediate flat regime from the farther CPTG outer attractor $V \propto r^{1/4}$. Fifth, dissociative cluster mergers should allow lensing support to separate from collisional gas when coherent directional transport is active.

These predictions make the structure sector falsifiable. CPTG would be weakened if high-acceleration systems failed to recover the baryonic limit, if weak-field galaxies did not show the predicted polarization scaling, if solved structural diagnostics were uncorrelated with residual organization, or if merger-plane transport could not reproduce gas-lensing dissociation without adding a dark matter halo.

10 Core Branch Parameter Inventory

This section lists the fixed quantities used by the core CPTG pi branch and separates them from the local structural quantities used in galaxy, compact-object, cluster, and redshift-stress reductions. Native geometric parameters define the theory-side branch. Transport, acoustic, clock, baryon, radiation, and lensing quantities are retained here only where they arise as reduced consequences of the same curvature-polarization and curvature-transport structure. When needed, these quantities can be translated into observational comparison coordinates, but their role here is to define the native theory-side branch.

The most important distinction is between the local structural scale $a_\star$ and the cosmic geometric anchor $a_{\pi,0}$. In galaxy and compact-structure equations, $a_\star$ is not a universal constant. It is the local acceleration scale associated with the structure being modeled. CPTG links acceleration to structure, so a galaxy, compact object, cluster, high-redshift system, and cosmic branch may carry different acceleration scales while still belonging to the same curvature-response family. The cosmic quantity $a_{\pi,0}$, by contrast, is the fixed native anchor of the geometric pi branch.

Class	Meaning	Examples
Fixed native branch quantities	Geometric quantities defining the locked CPTG branch	$p_C, a_{\pi,0}, \mathcal{B}_\pi, \mathcal{T}_\pi$
Derived branch quantities	Quantities calculated from the native branch rather than independently fitted	$H_\infty, H_0^{(\pi)}, R_C, \mathcal{G}_T$
Local structural quantities	System-dependent quantities tied to the curvature-response state of a bound structure	$a_\star, N, \Sigma, g_{\text{geom}}, \chi_{\text{grad}}$
Comparison-layer quantities	Observational-coordinate expressions of the native branch	$H_{0,\text{CMB}}^{\text{CPTG}}, \tau_{\text{eff}}, A_{\text{bandpower}}, S_{8,\text{late}}^{\text{CPTG}}$
Observed inputs	Measured quantities used to test a structural reduction	$M_b, R_e, \sigma_\star, V(R), z$

Table 1: CPTG parameter classes.

Quantity	Role	Status
$a_\star$	Local structural acceleration scale of the system being modeled	Regime-dependent structural scale
$a_{\pi,0}$	Fixed geometric acceleration anchor of the cosmic pi branch	Native branch quantity
N	Structural Mode controlling the curvature-response branch of a galaxy or bound system	Derived structural diagnostic
Σ	Transport or suppression factor controlling how strongly curvature response propagates through a system	Regime-dependent response variable
$g_{\text{bar,CPTG}}$	Baryonic source field used in the CPTG galaxy reduction	Computed from baryonic structure
g_{pol}	Curvature-polarization contribution to the reduced field	Solved response term
g_{geom}	Galaxy-limit geometric transport remnant	Reduced transport term
χ_{grad}	Compact-object curvature-gradient state	Structural closure variable
χ_{crit}	Threshold for finite-core compact-object closure	Theory threshold

Table 2: Local structural quantities used in CPTG reductions.

10.1 Native Geometric Parameters

Quantity	Definition	Value
p_C	Curvature exponent	π
$\mathcal{B}_\pi$	Late/geometric branch weight	$1 - 1/\pi = 0.681690113816$
$\mathcal{T}_\pi$	Transport branch weight	$1/\pi = 0.318309886184$
$a_{\pi,0}$	Geometric acceleration anchor	$5.568327996831708 \times 10^{-10} \text{ m s}^{-2}$
H_∞	Asymptotic Hubble scale	$57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}$
$H_0^{(\pi)}$	Raw geometric Hubble scale	$69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}$
R_C	Branch curvature scale	$788.7823091477 \text{ Gpc}$

Table 3: CPTG-native geometric parameters.

10.2 Transport, Acoustic, and Clock Parameters

Quantity	Definition	Value
p_{ac}	Acoustic exponent	$3 - \pi/100 = 2.968584073464$
$\mathcal{G}_T$	Transport factor	$p_{\text{ac}}/p_C = 0.944929658551$
$\sqrt{\mathcal{G}_T}$	Hubble-rate transport factor	0.972074924351
$\mathcal{G}_T^2$	Clock/tau transport gate	0.892892059610
$H_{0,\text{CMB}}^{\text{CPTG}}$	Transported Hubble comparison scale	$67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}$
$H_{\infty,\text{CMB}}^{\text{CPTG}}$	Transported asymptotic scale	$55.7127237837 \text{ km s}^{-1} \text{ Mpc}^{-1}$
$\Omega_{m,\text{ac}}$	Acoustic matter comparison fraction	0.314976552918
$\Omega_{\Lambda,\text{ac}}$	Complementary branch fraction	0.685023447082
$t_{0,\text{clock}}^{\text{CPTG}}$	Clock-branch age	13.88517358 Gyr

Table 4: Transport, acoustic, and clock parameters.

10.3 Baryon, Radiation, and Lensing Parameters

Quantity	Definition	Value
f_b	Baryon fraction	$\pi/20 = 0.157079632679$
$\Omega_b h_{\text{ac}}^2$	Acoustic baryon-density reduction	0.0225278574944
η_{10}^{ac}	Acoustic baryon-to-photon reduction	6.170380167710
$\Omega_b h_{\text{BBN}}^2$	Transport-scaled baryon conversion	0.021898765370
η_{10}^{BBN}	Transport-scaled baryon-to-photon conversion	5.998071834744
$N_{\text{eff}}^{\text{CPTG}}$	Effective radiation branch value	2.968584073464
A_L^{CPTG}	Native lensing response scale	$(p_C/p_{\text{ac}})^2 = 1.119956202138$
$A_{\text{bandpower}}$	Raw CMB-lensing bandpower conversion	1.009690662776
$S_{8,\text{late}}^{\text{CPTG}}$	Late-time reduced lensing amplitude	0.788071335232
$S_{8,\text{CMBL}}^{\text{CPTG}}$	Reduced CMB-lensing amplitude	0.821191065949

Table 5: Baryon, radiation, and lensing reduced branch quantities.

11 Background Geometry and Clock Branch

The CPTG background branch is defined by a geometric expansion function

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_C}. \quad (62)$$

The curvature exponent is

$$p_C = \pi. \quad (63)$$

The late-branch geometric weight is

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad (64)$$

and the transport-branch weight is

$$\mathcal{T}_\pi = \frac{1}{\pi}. \quad (65)$$

Therefore,

$$E_\pi(a)^2 = 1 - \frac{1}{\pi} + \frac{1}{\pi} a^{-\pi}. \quad (66)$$

In redshift form,

$$E_\pi(z)^2 = 1 - \frac{1}{\pi} + \frac{1}{\pi} (1+z)^\pi. \quad (67)$$

The raw geometric present-epoch normalization is

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}}, \quad (68)$$

with

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (69)$$

The acoustic and clock branch uses the transported exponent

$$p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (70)$$

so that

$$p_{\text{ac}} = 2.968584073464. \quad (71)$$

The corresponding clock-branch expansion function is

$$E_{\text{clock}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}. \quad (72)$$

Equivalently,

$$E_{\text{clock}}(a)^2 = 1 - \frac{1}{\pi} + \frac{1}{\pi} a^{-(3-\pi/100)}. \quad (73)$$

The transport factor between the curvature branch and the acoustic-clock branch is

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C}. \quad (74)$$

Numerically,

$$\mathcal{G}_T = 0.944929658551. \quad (75)$$

Hubble-like rates transport by

$$H_{\text{transported}} = H_{\text{raw}} \sqrt{\mathcal{G}_T}, \quad (76)$$

while distance-like and ruler-like quantities transport inversely:

$$D_{\text{transported}} = \frac{D_{\text{raw}}}{\sqrt{\mathcal{G}_T}}. \quad (77)$$

The transported CMB-comparison Hubble scale is therefore

$$H_{0,\text{CMB}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T}. \quad (78)$$

This gives

$$H_{0,\text{CMB}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (79)$$

The transported asymptotic scale is

$$H_{\infty, \text{CMB}}^{\text{CPTG}} = H_{\infty} \sqrt{\mathcal{G}_T}, \quad (80)$$

with

$$H_{\infty, \text{CMB}}^{\text{CPTG}} = 55.7127237837 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (81)$$

The clock age is defined by the clock-branch integral

$$t_0^{\text{CPTG}} = \frac{1}{H_{0, \text{CMB}}^{\text{CPTG}}} \int_0^1 \frac{da}{a E_{\text{clock}}(a)}. \quad (82)$$

Numerically,

$$t_0^{\text{CPTG}} = 13.88517358 \text{ Gyr}. \quad (83)$$

This age is a clock-branch comparison quantity. It is derived from the transported CPTG expansion law and the acoustic-clock exponent, not fitted as an independent age parameter.

12 Perturbation, Growth, and Lensing Channels

CPTG treats growth and lensing as reduced perturbation channels of the nonlinear curvature-polarization and curvature-transport field. In scalar perturbation form, the metric is written as

$$ds^2 = - (1 + 2\Psi) c^2 dt^2 + a(t)^2 (1 - 2\Phi) d\mathbf{x}^2. \quad (84)$$

The growth response is defined by the effective Poisson relation

$$k^2 \Psi = -4\pi G a^2 \bar{\rho} \mu_{\text{CPTG}}(a, k) \delta. \quad (85)$$

The lensing response is defined through the Weyl-potential combination

$$k^2 (\Phi + \Psi) = -8\pi G a^2 \bar{\rho} \Sigma_{\text{CPTG}}(a, k) \delta. \quad (86)$$

For the reduced comparison layer used here, the active scale-averaged growth and lensing responses are

$$\mu_{\text{CPTG}}(a) = 1 + \frac{\pi}{2} (A_L^{\text{CPTG}} - 1) (1 - a^{p_{\text{ac}}}), \quad (87)$$

$$\Sigma_{\text{CPTG}}(a) = 1 + (A_L^{\text{CPTG}} - 1) (1 - a^{p_{\text{ac}}}). \quad (88)$$

The native lensing response scale is fixed by the curvature-to-acoustic exponent ratio:

$$A_L^{\text{CPTG}} = \left(\frac{p_C}{p_{\text{ac}}} \right)^2. \quad (89)$$

With

$$p_C = \pi \quad (90)$$

and

$$p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (91)$$

this gives

$$A_L^{\text{CPTG}} = 1.119956202138. \quad (92)$$

The raw CMB-lensing bandpower comparison is not the same object as the native lensing response. The projected bandpower amplitude is

$$A_{\text{bandpower}} = 1 + q_{\text{bandpower}} (A_L^{\text{CPTG}} - 1), \quad (93)$$

where

$$q_{\text{bandpower}} = \frac{1 - \mathcal{G}_T}{\mathcal{B}_\pi}. \quad (94)$$

Therefore,

$$A_{\text{bandpower}} = 1.009690662776. \quad (95)$$

The late-time weak-lensing comparison amplitude is

$$S_{8,\text{late}}^{\text{CPTG}} = 0.788071335232. \quad (96)$$

The compressed CMB-lensing amplitude channel is

$$S_{8,\text{CMBL}}^{\text{CPTG}} = S_{8,\text{late}}^{\text{CPTG}} (A_L^{\text{CPTG}})^{\alpha_C}. \quad (97)$$

The compression exponent is

$$\alpha_C = 1 - \frac{2}{\pi}. \quad (98)$$

This gives

$$S_{8,\text{CMBL}}^{\text{CPTG}} = 0.821191065949. \quad (99)$$

Thus CPTG distinguishes three lensing-related quantities:

$$A_L^{\text{CPTG}}, \quad A_{\text{bandpower}}, \quad S_{8,\text{CMBL}}^{\text{CPTG}}. \quad (100)$$

The first is the native response scale, the second is a projected raw bandpower comparison, and the third is a compressed amplitude comparison. These are different observables and should not be collapsed into a single fitted lensing parameter.

13 Geometric Origin of the CMB Amplitude and Visibility Gates

The locked CMB comparison layer contains two geometric projection gates beyond the cold-sector distance map: the scalar-amplitude gate and the optical-depth visibility gate. These are not fitted parameters, and they are not borrowed from an external cosmological model. They are CPTG curvature projections that translate native curvature transport into the acoustic amplitude and visibility coordinates measured by the CMB sky.

The governing curvature branch is

$$p_C = \pi. \quad (101)$$

The acoustic transport branch is

$$p_{ac} = 3 - \frac{\pi}{100}. \quad (102)$$

The corresponding acoustic-to-curvature transport ratio is

$$\mathcal{G}_T = \frac{p_{ac}}{p_C}. \quad (103)$$

The purpose of the comparison layer is therefore narrow: CPTG-native curvature quantities are projected into acoustic observables, visibility suppression, and lensing reconstruction coordinates. The theory remains CPTG geometry throughout the mapping.

The gate closures in this section should therefore be read as reduced geometric closure rules, not as numerical patch terms. The amplitude gate fixes how the native curvature branch is projected into a scalar acoustic-amplitude coordinate. The visibility gate fixes how acoustic transport suppresses the observed scalar amplitude through the optical-depth coordinate. Once these maps are fixed, the CMB comparison row is a consequence of the branch rather than a sector-by-sector retuning.

13.1 Amplitude Projection and the $\pi/3$ Curvature Gate

The scalar amplitude in the CMB comparison layer is the two-point scalar projection of a CPTG curvature-response field. The native curvature branch is organized by $p_C = \pi$, while the acoustic scalar normalization is a three-spatial-direction projection. The branch-to-acoustic projection is therefore

$$\mathcal{A}_C = \frac{p_C}{3}. \quad (104)$$

This closure says that the scalar acoustic amplitude is not normalized by a free phenomenological amplitude. It is the three-direction acoustic projection of the

curvature branch itself. The numerator carries the native curvature exponent; the denominator carries the spatial acoustic projection dimension.

Since

$$p_C = \pi, \quad (105)$$

the amplitude projection becomes

$$\mathcal{A}_C = \frac{\pi}{3}. \quad (106)$$

The effective scalar amplitude used in the CMB comparison layer is therefore

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \mathcal{A}_C. \quad (107)$$

Substituting the geometric amplitude projection gives

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \frac{\pi}{3}. \quad (108)$$

For the locked CPTG-native amplitude,

$$A_{s,\text{CPTG}} = 2.04 \times 10^{-9}, \quad (109)$$

the projected comparison-layer amplitude is

$$A_{s,\text{eff}} = 2.04 \times 10^{-9} \frac{\pi}{3}. \quad (110)$$

Thus,

$$A_{s,\text{eff}} = 2.136283004441 \times 10^{-9}. \quad (111)$$

This value is not introduced as an independent normalization. It is the scalar CMB projection of the CPTG curvature-amplitude branch.

13.2 Visibility Transport and the $\mathcal{G}_T^2$ Optical-Depth Gate

The optical-depth comparison coordinate measures visibility-layer suppression of the observed CMB amplitude. In CPTG, this suppression is a curvature-transport effect. It is governed by the same acoustic-to-curvature transport ratio

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C}. \quad (112)$$

The visibility projection contains two geometric contractions. The first maps the curvature branch into the acoustic transport branch. The second maps the acoustic visibility layer into the scalar comparison coordinate. Therefore the visibility

projection is quadratic in the transport ratio. In reduced form, optical depth is not being adjusted to rescue the amplitude; it is the scalar visibility expression of the same acoustic-to-curvature transport ratio that defines the branch:

$$\mathcal{T}_\tau = \mathcal{G}_T^2. \quad (113)$$

The effective optical depth in the CMB comparison layer is then

$$\tau_{\text{eff}} = \tau_{\text{CPTG}} \mathcal{T}_\tau. \quad (114)$$

Using the CPTG visibility projection,

$$\tau_{\text{eff}} = \tau_{\text{CPTG}} \mathcal{G}_T^2. \quad (115)$$

For the locked native value

$$\tau_{\text{CPTG}} = 0.066, \quad (116)$$

and

$$\mathcal{G}_T^2 = 0.892892059610, \quad (117)$$

the projected comparison-layer optical depth is

$$\tau_{\text{eff}} = 0.066 \times 0.892892059610. \quad (118)$$

Thus,

$$\tau_{\text{eff}} = 0.058930875934. \quad (119)$$

The optical-depth gate is therefore the visibility-layer expression of CPTG acoustic transport. It is not an empirical retuning of optical depth.

13.3 Locked Gate Closure

The CMB comparison row closes through three CPTG geometric maps:

$$\omega_{c,\text{eff}} h^2 = \omega_{c,\text{CPTG}} h^2 \mathcal{D}_c, \quad (120)$$

$$A_{s,\text{eff}} = A_{s,\text{CPTG}} \frac{\pi}{3}, \quad (121)$$

$$\tau_{\text{eff}} = \tau_{\text{CPTG}} \mathcal{G}_T^2. \quad (122)$$

The curvature-distance gate $\mathcal{D}_c$ is already fixed by the CPTG comparison geometry. The amplitude and visibility gates now have direct geometric interpretations:

$\pi/3$ is the scalar acoustic projection of the π -curvature branch, while $\mathcal{G}_T^2$ is the two-step visibility contraction of the acoustic transport branch.

The locked effective values are therefore

$$\omega_{c,\text{eff}} h^2 = 0.117685841620526, \quad (123)$$

$$A_{s,\text{eff}} = 2.136283004441 \times 10^{-9}, \quad (124)$$

$$\tau_{\text{eff}} = 0.058930875934. \quad (125)$$

The lensing amplitude remains fixed at

$$A_{\text{lens}} = 1. \quad (126)$$

No lensing inflation is introduced. No posterior retuning is introduced. The locked CMB comparison layer is therefore a geometric projection of CPTG curvature transport into acoustic amplitude, visibility, matter-density, and lensing coordinates.

14 Scalable CMB Comparison-Layer Capability

CPTG can translate its native curvature branch into CMB-facing comparison layers because the same geometric quantities control acoustic transport, scalar-amplitude projection, optical-depth visibility, matter-density projection, and lensing response. The branch is not rebuilt separately for each CMB likelihood. Instead, the native curvature scale, acoustic transport ratio, and visibility contraction define a fixed map from CPTG geometry into the conventional coordinates used to compare with CMB data.

This scalability is part of the theory rather than a bookkeeping device. A single curvature-response branch supplies the expansion scale, the acoustic branch, the cold-sector projection, the scalar-amplitude gate, the visibility gate, and the lensing normalization. Once those maps are fixed, CMB comparisons become tests of whether one geometric branch can be expressed consistently across temperature, polarization, lensing, and combined-stack summaries without introducing a separate lensing inflation parameter or retuning the branch for each dataset.

15 Prediction Families and Cross-Regime Falsifiability

The reductions developed above give CPTG a prediction layer that is broader than any one observational channel. The purpose of this section is not to list isolated targets or object-specific case studies. It is to state the main theory-level prediction families that follow when the same curvature-polarization variables, transport variables, structural modes, and expansion branch are carried across galaxies, clusters, compact structures, redshift stress, effective-radius support, growth, and observational comparison layers.

A prediction is strongest when the same geometric variables constrain more than one regime. CPTG is therefore falsifiable not only by whether it can recover a single class of data, but by whether the same curvature-polarization and transport framework remains mutually consistent across structurally different systems.

15.1 Compact Objects and Finite-Core Curvature Closure

In high-gradient systems CPTG predicts that compact-object formation is governed by curvature organization as well as mass [9]. The relevant invariant combines the field-strength sector and the field-gradient sector,

$$C = g_j g_j + \kappa^2 (\partial_i g_j) (\partial_i g_j), \quad (127)$$

with dimensionless response variables

$$\chi = \frac{C}{a_\star^2}, \quad (128)$$

$$\chi_{\text{grad}} = \frac{\kappa^2 (\partial_i g_j) (\partial_i g_j)}{a_\star^2}. \quad (129)$$

The polarization sector may enter through a high-gradient response of the form

$$\mathcal{L}_{\text{pol}} = \beta a_\star^2 \chi^{1/3}. \quad (130)$$

Because the response is organized by finite curvature-gradient structure, CPTG does not admit an infinite-core singularity as a mathematical endpoint. Compact-object closure is structural rather than mass-only,

$$\chi_{\text{grad}} \geq \chi_{\text{crit}}, \quad (131)$$

and compact objects are classified by both physical core mass and gradient state,

$$(M_{\text{core}}, \chi_{\text{grad}}). \quad (132)$$

The interior branch is finite-core and regular,

$$\Phi(r) = \Phi_0 + \alpha_{\text{core}} r^2 + O(r^4), \quad (133)$$

$$g(r) \simeq -2\alpha_{\text{core}} r, \quad (134)$$

so that

$$\lim_{r \rightarrow 0} g(r) = 0. \quad (135)$$

Thus CPTG predicts compact objects with finite-core curvature closure, nonzero structural tidal response, and exterior curvature polarization that can make the apparent gravitational mass differ from the physical core mass,

$$M_{\text{apparent}} = A_{\text{pol}} M_{\text{core}}. \quad (136)$$

The exterior field should interpolate between a near-field branch and a structural far-field branch,

$$g_{\text{near}}(r) \simeq \frac{GM}{r^2}, \quad (137)$$

$$g_{\text{far}}(r) = g_0 \left(\frac{r_0}{r} \right)^{\nu_N}, \quad (138)$$

$$\nu_N = \nu(N, \chi, \Sigma). \quad (139)$$

Observable compact-object quantities should therefore be linked through common parameters. Apparent mass amplification, finite-core tidal response, ejecta energetics, transport suppression, remnant scale, and possible echo structure should not be independently tunable.

The transport sector also predicts suppression in high-gradient interiors through a schematic term of the form

$$\mathcal{L}_{\text{trans}} \propto -(S^\alpha \Sigma \nabla_\alpha \Phi)^2, \quad (140)$$

with

$$S^\alpha S_\alpha = \pm 1, \quad (141)$$

$$0 \leq \Sigma \leq 1, \quad (142)$$

$$\Sigma \rightarrow 0 \quad \text{as} \quad \chi_{\text{grad}} \rightarrow \chi_{\text{crit}} \quad \text{or} \quad \chi \rightarrow \infty. \quad (143)$$

15.2 Structural Collapse and Vanishing-Source Channels

The same finite-core closure gives a structural-collapse prediction family. CPTG does not classify collapse only by enclosed mass. It classifies the event by physical core mass, gradient state, apparent polarization amplification, geometric mass contribution, and transport suppression [10],

$$\mathcal{S}_{\text{CPTG}} = \{M_{\text{core}}, \chi_{\text{grad}}, A_{\text{pol}}, M_{\text{geom}}, \Sigma_{\text{trans}}\}. \quad (144)$$

A quiet or vanishing-source channel is favored when the physical core is smaller than the apparent gravitational mass,

$$M_{\text{core}} < M_{\text{apparent}}, \quad (145)$$

$$A_{\text{pol}} > 1, \quad (146)$$

$$\chi_{\text{grad}} \rightarrow \chi_{\text{crit}}, \quad (147)$$

and when electromagnetic recovery is transport-suppressed,

$$\frac{L}{L_{\text{prog}}} \ll 1, \quad (148)$$

$$\frac{L_X}{L_{\text{bol}}} \ll 1. \quad (149)$$

The falsifiable content is that apparent mass, ejecta scale, remnant luminosity, X-ray weakness, and late-time infrared or envelope behavior should trace the same structural state rather than vary independently.

15.3 Compact-Structure and Redshift-Stress Predictions

At high redshift CPTG predicts that compact stellar systems probe the relation between local support scale and cosmic branch state [11]. For a compact support radius R_c ,

$$a_\sigma(R_c) = \frac{\sigma_\star^2}{R_c}, \quad (150)$$

$$E(z) = \frac{H(z)}{H_0}, \quad (151)$$

$$a_{\sigma,z}(R_c) = \frac{a_\sigma(R_c)}{E(z)}. \quad (152)$$

The time-linked branches are

$$a_{\star,0}(R_c) = a_{\sigma,z}(R_c), \quad (153)$$

$$a_\star(t_{\text{em}}; R_c) = a_{\star,0}(R_c)E(z) = a_\sigma(R_c). \quad (154)$$

The corresponding CPTG curvature radius is

$$R_C(t) = \frac{48\pi c^2}{a_\star(t)}. \quad (155)$$

Here R_c denotes the compact support scale, while R_C denotes the CPTG curvature radius. The prediction is a branch relation over radius and redshift: larger support radii lower the normalized stress scale, while earlier emission epochs map the same local support into a different cosmic curvature branch.

15.4 Rotation-Aware and Jeans-Supported Structural Tests

CPTG predicts that systems with both ordered rotation and random stellar motion require a rotation-aware effective-radius channel rather than a pressure-only reduction. At an effective radius,

$$V_{\text{rms}}^2(R_e) = \sigma_0^2 + V_{R_e}^2, \quad (156)$$

$$a_{\text{rms}}(R_e) = \frac{V_{\text{rms}}^2(R_e)}{R_e}, \quad (157)$$

$$a_\sigma = \frac{\sigma_0^2}{R_e}. \quad (158)$$

A reduced baryonic field and polarization response can be written as

$$g_{\star}(R_e) = \frac{GM_{\star}}{R_e^2}, \quad (159)$$

$$\mathcal{R}_{\text{PT}}(R_e) = \left[\frac{g_{\star}(R_e)}{a_{\sigma}} \right]^{1/2}, \quad (160)$$

$$g_{\text{CPTG}}(R_e) = \mathcal{R}_{\text{PT}}(R_e)g_{\star}(R_e). \quad (161)$$

The recovered rotation component is then

$$V_{\text{CPTG}}(R_e) = \sqrt{g_{\text{CPTG}}(R_e)R_e - \sigma_0^2}, \quad (162)$$

with effective-radius structural mode

$$N_{R_e} = \frac{g_{\text{CPTG}}(R_e)}{a_{\sigma}}. \quad (163)$$

The falsifiable distinction is direct: pressure-only reductions should undercount support when ordered rotation is dynamically important.

For fully dispersion-supported systems, the separate Jeans-supported channel is controlled by the radial Jeans field,

$$g_{\text{Jeans}}(r) = -\frac{\sigma_r^2(r)}{r} \left[\frac{d \ln \rho_{\star}}{d \ln r} + \frac{d \ln \sigma_r^2}{d \ln r} + 2\beta_J(r) \right], \quad (164)$$

where

$$\beta_J(r) = 1 - \frac{\sigma_{\theta}^2(r)}{\sigma_r^2(r)}. \quad (165)$$

An effective-radius spheroid proxy may be written as

$$g_{\star}(R_e) = \frac{GM_{\star}}{(R_e + a_H)^2}, \quad (166)$$

$$a_H = \frac{R_e}{1.8153}. \quad (167)$$

In this channel the CPTG pressure-supported prediction is summarized by

$$\mathcal{R}_{\text{PT}}(R_e) = \frac{g_{\text{CPTG}}(R_e)}{g_{\star}(R_e)}, \quad (168)$$

$$\sigma_{\text{CPTG}}(R_e) = \sigma_{\text{raw}}(R_e) \sqrt{\mathcal{R}_{\text{PT}}(R_e)}, \quad (169)$$

$$N_{R_e} = \frac{g_{\text{CPTG}}(R_e)}{a_{\star}}. \quad (170)$$

A full radial validation requires resolved density, dispersion, anisotropy, geometry, projection, and lensing information. Without those inputs, the effective-radius version remains a reduced structural prediction rather than a complete Jeans solution.

15.5 Cosmological Scale Relation and Growth Predictions

CPTG treats the local structural acceleration scale and the cosmological Hubble scale as different regime expressions of curvature organization. The dimensional bridge is

$$H_{\star} = \frac{a_{\star}}{c}, \quad (171)$$

but CPTG does not require the identity

$$H_{\star} = H_0. \quad (172)$$

Instead,

$$a_{\star} = \mathcal{A}(\mathcal{S}_{\text{local}}), \quad (173)$$

$$a_{\star} = \mathcal{A}(\mathcal{S}_{\text{local}}, \rho_b, R, \mathcal{R}_{\text{PT}}), \quad (174)$$

$$H_0 = \mathcal{H}(\mathcal{S}_{\text{cos}}), \quad (175)$$

$$H(z) = \mathcal{H}(\mathcal{S}_{\text{cos}}(z)). \quad (176)$$

A redshift evolution channel may be written as

$$a_{\star}(z) = a_{\star}(0) \frac{H(z)}{H_0}, \quad (177)$$

or more generally as

$$a_{\star}(z) = \mathcal{A}(\mathcal{S}_{\text{local}}(z), \mathcal{S}_{\text{cos}}(z)). \quad (178)$$

The cosmic-web state also changes with time,

$$\mathcal{S}_{\text{early}} \simeq \mathcal{S}_{\text{uniform}} + \delta\mathcal{S}, \quad (179)$$

$$\mathcal{S}_{\text{late}} = \mathcal{S}_{\text{void}} + \mathcal{S}_{\text{filament}} + \mathcal{S}_{\text{sheet}} + \mathcal{S}_{\text{halo}}. \quad (180)$$

Environmental screening can be represented by

$$a_{\star, \text{eff}} = \mathcal{E}_{\text{env}} a_{\star}, \quad (181)$$

$$0 < \mathcal{E}_{\text{env}} \leq 1, \quad (182)$$

$$Q_g = (\partial_i g_j)(\partial_i g_j). \quad (183)$$

The resulting prediction is stronger than a loose analogy between local and cosmic scales. CPTG predicts that distance-ladder residuals, growth residuals, structural modes, redshift-stress behavior, and environmental screening should correlate with curvature-gradient organization. They are different observational reductions of the same geometric response, not unrelated sector-by-sector anomalies.

15.6 Expansion-Branch and Comparison-Layer Predictions

The native expansion branch remains the geometric pi branch,

$$E_\pi(a)^2 = \left(1 - \frac{1}{\pi}\right) + \frac{1}{\pi}a^{-\pi}, \quad (184)$$

with

$$H_{\text{geom}}(a) = H_0^{(\pi)} E_\pi(a). \quad (185)$$

The generic comparison-layer prediction is that different observational channels measure projected expressions of the same native branch,

$$H_i(a) = \mathcal{P}_i(a) \mathcal{T}_i(a) H_{\text{geom}}(a), \quad (186)$$

where $\mathcal{P}_i$ and $\mathcal{T}_i$ encode channel-specific polarization and transport response. In the acoustic comparison layer,

$$H_{0,\text{CMB}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T}, \quad (187)$$

while in a local luminosity-distance projection,

$$H_{0,\text{loc}}^{\text{CPTG}} = H_0^{(\pi)} \mathcal{L}_{\text{loc}}. \quad (188)$$

Thus the expansion prediction is not two incompatible expansion histories. It is a two-sided projection structure around one native CPTG branch. The local, acoustic, lensing, abundance, and large-scale-structure comparison rows should become mutually constrained once the polarization and transport operators are derived rather than assigned.

15.7 Cross-Regime Falsifiability

The central falsifiability requirement is that CPTG cannot tune each regime independently. The same curvature-polarization scale, transport suppression, structural mode, convergence behavior, compact-object gradient state, and expansion projection operators must remain mutually consistent. A galaxy structural mode should connect to outer-slope behavior. A cluster-merger transport reduction should connect to displaced lensing. A compact-object closure should connect finite-core response, apparent mass, ejecta scale, and transport suppression. A high-redshift compact structure should connect support scale, redshift-normalized stress, and curvature radius. A CMB comparison row should connect to growth, lensing, baryon conversion, and expansion projection without requiring an inflated lensing amplitude.

This makes CPTG predictive in the strict sense: success in one sector increases the constraint on the others. A failure in any cleanly measured regime is not

hidden by introducing a new independent component. It becomes a stress test of the shared geometric branch. That is the intended standard for the theory: not isolated agreement with selected observables, but cross-regime recovery from one curvature polarization and transport framework.

16 Conclusion

Curvature Polarization Transport Gravity is presented here as a unified geometric framework in which cosmic structure and expansion arise from a common curvature-response branch. The construction does not begin with a dark halo, an interpolation function, or an observational fit coordinate. It begins with a native curvature-polarization and curvature-transport structure organized by

$$p_C = \pi, \quad (189)$$

$$p_{ac} = 3 - \frac{\pi}{100}, \quad (190)$$

$$\mathcal{G}_T = \frac{p_{ac}}{p_C}, \quad (191)$$

and anchored by the geometric acceleration scale

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2}. \quad (192)$$

The corresponding expansion branch is

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_C}, \quad (193)$$

while the acoustic and clock branches follow from the transported exponent. In this form, expansion, clock transport, growth response, lensing response, baryon conversion, and CMB-facing comparison quantities are not separate fitted sectors. They are reduced scalar expressions of the same geometric branch.

The galaxy reduction gives the theory its most direct structural content. From a single nonlinear curvature-polarization acceleration equation, CPTG produces the baryonic Tully–Fisher amplitude scaling, a finite-domain Structural Mode N , and a far-outer Slope of Convergence. These are different readings of the same solved field. The Tully–Fisher relation captures the leading amplitude law of the weak-field polarization regime. Structural Mode N captures the organization of the solved curvature support across the resolved galaxy. The Slope of Convergence captures the continuation behavior of the same equation beyond the measured rotation-curve domain. This gives CPTG a structural prediction layer that is not merely a curve fit: the theory assigns galaxies an internally computed curvature-organization observable and an outer-domain convergence behavior.

The cluster-merger reduction extends this same framework into systems where directional transport is essential. In dissociative mergers, the baryonic gas, galaxy

distribution, transported curvature mode, and lensing response need not remain coincident. CPTG therefore treats displaced lensing structure as a transport-and-polarization effect of the curvature field rather than as a separate collisionless matter component inserted into the system.

The cosmological branch completes the same picture at large scale. The geometric pi branch defines the native expansion law, the acoustic exponent defines the transported clock branch, and the scalar comparison maps express how the same CPTG-native quantities can be translated into CMB-related acoustic, matter-density, amplitude, visibility, and lensing coordinates. This comparison-layer capability is scalable because it follows from the branch structure itself. It does not redefine CPTG in conventional observational-parameter language; it expresses the native theory in the coordinates required for contact with CMB, abundance, distance, growth, and large-scale-structure summaries.

The central claim of this paper is therefore theoretical. CPTG supplies one geometric framework in which galaxy rotation support, structural mode organization, outer-slope convergence, dissociative cluster lensing transport, cosmic expansion, acoustic clock transport, growth, lensing, and CMB-facing comparison layers are connected through curvature polarization and curvature transport. The remaining task is not to replace this native structure with separate sector-by-sector parameter fits, but to continue deriving, testing, and tightening the reduced limits of one scalable geometric theory.

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